

Signal Processing

Exercise 3

SoSe26

Question 1

Determine if the given systems $\mathfrak{S}(x(t))$: $x(t) \rightarrow y(t)$ are linear, causal and time-invariant. Please make sure that you briefly explain/prove each of your conclusion.

System 1.1	$y(t) = \dot{x}(t)$	Differentiation
System 1.2	$y(t) = \dot{x}(t) + 2$	Differentiation
System 1.3	$y(t) = x^2(t)$	Exponentiation
System 1.4	$y(t) = x(t) \sin(\omega_0 t)$	Amplitude Modulation

Exam preparation: Determine if the given system $\mathfrak{S}(x(t))$: $x(t) \rightarrow y(t)$ are linear, causal and time-invariant. Please make sure that you briefly explain/prove each of your conclusion.

System 1.5	$y(t) = \int_{-\infty}^{t-1} x(\tau) d\tau$	Integration
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